PLANTCARE AI

Team Work Documentation & Project Overview

Команда проекта (Roles)

- **PM:** Isakova Aidena
- **Frontend:** Kadyrkulova Amina
- **Backend:** Bazarbekov Almazbek
- **AI:** Mamadzhanova Amina
- **DevSecOps:** Anarbaeva Asel

1. Project Overview (Обзор проекта)

Project name: PlantCare

Project description: PlantCare AI is an AI-powered web application that helps users identify plants from photos, analyze their condition, and receive personalized care recommendations.

Problem: Many people have home plants but do not know how to care for them properly. They may forget to water plants, overwater them, place them in poor lighting conditions, or fail to understand why leaves turn yellow, dry, or damaged.

Solution: Our web application allows users to upload a photo of their plant and fill in basic information such as city, indoor/outdoor location, and sunlight level. The AI analyzes the image and provides the plant name, condition, possible issues, watering advice, light advice, fertilizing advice, and a care schedule.

Main MVP goal: The main goal of our MVP is to build a working flow: Upload plant photo → AI analysis → Result page with care recommendations.

2. Team Members and Roles (Роли и обязанности)

Our team works together using Jira, GitHub, daily standups, and shared documentation. Each team member has their own role and responsibility, but the project is developed as one connected product, not as separate individual parts.

Team member	Role	Main responsibility
Aidena	Project Manager	Organizes the team workflow, manages Jira, defines MVP scope, documents standups, tracks blockers, and prepares the presentation structure.
Amina K.	Frontend Developer	Builds the user interface: landing page, Add Plant form, photo upload UI, AI result page, and connects frontend with backend API.
Almaz	Backend Developer	Creates the backend server, API endpoints, database structure, and connects the AI analysis flow with the frontend.
Amina M.	AI Engineer	Creates the AI prompt, defines the JSON response format, tests plant photo analysis, and improves AI output quality.
Asel	DevSecOps Engineer	Manages GitHub repository structure, Docker setup, environment variables, API key protection, and deployment.

4. Development (Разработка)

Each team member worked on their role:

- Frontend built the interface.
- Backend created API endpoints.
- AI Engineer created and tested AI prompt.
- DevSecOps prepared GitHub, Docker, and environment variables.
- Project Manager tracked tasks, blockers, and documentation.

5. Testing and Integration (Тестирование и интеграция)

We tested the full MVP flow: **Upload plant photo → AI analysis → Result page with care recommendations**

We checked:

- photo upload;
- backend response;
- AI JSON result;
- frontend display;
- loading and error messages;
- final demo flow.

Main Result: The project lifecycle helped our team move step by step from planning to a working MVP, instead of building separate parts without coordination.

9. Presentation Plan (План презентации)

The final presentation will show our problem, solution, MVP, team roles, tech stack, and demo.

Slide 1 — Project Name

Speaker: Project Manager

PlantCare AI — AI-powered plant care assistant.

Slide 2 — Problem

Speaker: Project Manager

Many people do not know how to care for home plants properly. They forget watering, overwater plants, or do not understand why plants become yellow, dry, or unhealthy.

Slide 3 — Solution

Speaker: AI Engineer

PlantCare AI allows users to upload a plant photo, analyze it with AI, and receive care recommendations and a simple care schedule.

Slide 4 — MVP Features

Speaker: AI Engineer

Our MVP includes: photo upload; AI plant analysis; result page; watering advice; light advice; fertilizing advice; care schedule.

Slide 5 — Demo

Speaker: Frontend Developer + Backend Developer

We will show the main user flow: Upload plant photo → AI analysis → Result page with care recommendations

Slide 6 — Tech Stack

Speaker: Backend Developer / DevSecOps Engineer

We used: React.js, Node.js + Express.js, PostgreSQL, AI Vision API, Docker, GitHub, Jira

Slide 7 — Team Roles

Speaker: All team members

- Project Manager — Jira, planning, documentation, blockers, presentation
- Frontend Developer — UI, forms, result page
- Backend Developer — API, server, database
- AI Engineer — AI prompt, JSON response, plant analysis
- DevSecOps Engineer — GitHub, Docker, .env, deployment

Slide 8 — Future Improvements

Speaker: DevSecOps Engineer

In the future, we can add: user accounts; weather-based recommendations; email reminders; advanced dashboard; plant disease detection; mobile app.

Main Presentation Goal: Our goal is to show that PlantCare AI is a small but working MVP with a clear problem, useful AI feature, and real teamwork behind it.